Binomial Probability Distribution

$$P(x=k)=C_k^n p^k q^{n-k}=\frac{n!}{k!(n-k)!}p^k q^{n-k}$$

Mean : $\mu = np$

Variance : $\sigma^2 = npq$

Standard deviation : $\sigma = \sqrt{npq}$

Hypergeometric Probability Distributio

$$P(x=k)=\frac{C_k^M C_{n-k}^{N-M}}{C_n^N}$$

M → successes, ***N-M*** → failures,

n → size of the random sample space

Mean : $\mu = n\left(\frac{M}{N}\right)$

Variance : $\sigma^2 = n\left(\frac{M}{N}\right)\left(\frac{N-M}{N}\right)\left(\frac{N-n}{N-1}\right)$

Poisson Probability Distribution

$$P(x=k)=\frac{\mu^k e^{-\mu}}{k!}$$

Mean: $E(x) = \mu$

Variance : $\sigma^2 = \mu$

Standard deviation: $\sigma = \sqrt{\mu}$

Variance of a Sample

$$s^2=\frac{\sum(x_i-\bar{x})^2}{n-1}=\frac{\sum x_i{}^2-\frac{(\sum x_i)^2}{n}}{n-1}$$

Variance of Population: $\sigma^2=\frac{\sum(x_i-\mu)^2}{N}$

Correlation coefficient, $r=\frac{s_{xy}}{s_x s_y}$

Normal Distribution, *N(μ, σ^2)*

Standardizing the value of $\bar{x}$:

$$z=\frac{\bar{x}-\mu}{\sigma/\sqrt{n}}$$

Standardizing the value $\hat{p}$:

$$z=\frac{\hat{p}-p}{\sqrt{\frac{pq}{n}}}$$

Statistical Process Control

$$\text{LCL}:\bar{\bar{x}}-3\frac{s}{\sqrt{n}} \quad \text{UCL}:\bar{\bar{x}}+3\frac{s}{\sqrt{n}}$$

$$\text{LCL}:\bar{p}-3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} \quad \text{UCL}:\bar{p}+3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$$

Confidence interval for a population mean μ :

$$\bar{x}\pm z_{\alpha/2}\frac{s}{\sqrt{n}}$$

Confidence interval for a population proportion p :

$$\hat{p}\pm z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n}}$$

Point estimator of population mean μ : $\bar{x}$

The 95% margin of error $(n \geq 30)$: $\pm 1.96\frac{s}{\sqrt{n}}$

Point estimator of population proportion p : $\hat{p} = x/n$

The 95% margin of error $(n \geq 30)$: $\pm 1.96\sqrt{\frac{\hat{p}\hat{q}}{n}}$

Standardizing the value of a test statistic :

$$\begin{Bmatrix} z \\ t \end{Bmatrix} = \frac{(test_statistic) - (population\ parameter)}{SE}$$

Standard Error (SE)

$SE = \frac{\sigma}{\sqrt{n}}$, or

$SE = \sqrt{\frac{\hat{p}\hat{q}}{n}}$, or

$SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$, or

$SE = \sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}$, or

$SE = \sqrt{s^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$ (for two small samples with common variance)

Parameter	Test Statistic
μ	$z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$
p	$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0 q_0}{n}}}$
$\mu_1 - \mu_2$	$z = \frac{(\bar{x}_1 - \bar{x}_2) - D_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$
$p_1 - p_2$	$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}\hat{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$ or $z = \frac{(\hat{p}_1 - \hat{p}_2) - D_0}{\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}}$

Bound, B

$z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right) \prec B$, or

$z_{\alpha}\left(\frac{\sigma}{\sqrt{n}}\right) \prec B$, or

$z_{\alpha/2}\left(\sqrt{\frac{pq}{n}}\right) \prec B$, or

$z_{\alpha}\left(\sqrt{\frac{pq}{n}}\right) \prec B$

pooled estimate for the common value of p (large samples)

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$$

Common variance for two samples (small samples)

$$s^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2}$$

$$F = \frac{s_1^2}{s_2^2}$$

Variance of a Sample

$$s^2 = \frac{\sum(x_i - \bar{x})^2}{n-1} = \frac{\sum x_i^{\,2} - \frac{(\sum x_i)^2}{n}}{n-1}$$

Variance of Population:

$$\sigma^2 = \frac{\sum(x_i - \mu)^2}{N}$$

Confidence Interval for σ^2

$$\frac{(n-1)s^2}{\chi^2_{\propto/2}} < \sigma^2 < \frac{(n-1)s^2}{\chi^2_{(1-\propto/2)}}$$

Confidence Interval for $\sigma_1^2 \ / \sigma_2^2$

$$\left(\frac{s_1^2}{s_2^2}\right)\frac{1}{F_{df_1,df_2}} < \frac{\sigma_1^2}{\sigma_2^2} < \left(\frac{s_1^2}{s_2^2}\right)F_{df_2,df_1}$$